

Graph NP-Completeness Proofs

Compact Actual Proofs - Exam Ready

Proof Format to Use

For every NP-completeness proof, keep this order:

$\boxed{\text{In NP}} \rightarrow \boxed{\text{Build}} \rightarrow \boxed{\Rightarrow} \rightarrow \boxed{\Leftarrow} \rightarrow \boxed{\text{Time}} \rightarrow \boxed{\text{Conclusion}}$

Reduction Map

$\text{CLIQUE} \leq_p \text{INDEPENDENT-SET}$, $3\text{-COLOR} \leq_p \text{VERTEX-COLOR}$, $\text{VERTEX-COLOR} \leq_p \text{CLIQUE-PARTITION}$

Also: $3\text{-CNF-SAT} \leq_p 3\text{-COLOR}$. For edge coloring, use Holyer's theorem.

1 Independent Set is NP-Complete

Problem

Does graph G contain at least k vertices with no edge between any two of them?

Reduction

$\boxed{\text{CLIQUE} \leq_p \text{INDEPENDENT-SET}}$

Given (G, k) , build the complement graph $\overline{G}$ and keep the same k .

Actual Proof

In NP. Certificate = k vertices. Check every chosen pair; none may be adjacent.

Build. In $\overline{G}$, two distinct vertices are adjacent exactly when they are *not* adjacent in G .

$\Rightarrow$. If S is a k -clique in G , every pair in S is an edge of G . Hence no pair is an edge of $\overline{G}$. So S is an independent set in $\overline{G}$.

$\Leftarrow$. If S is an independent set of size k in $\overline{G}$, no pair in S is an edge of $\overline{G}$. Therefore every pair is an edge of G , so S is a k -clique in G .

Time. Build $\overline{G}$ by checking all vertex pairs: $O(|V|^2)$.

Conclusion.

$\boxed{\text{INDEPENDENT-SET is NP-complete.}}$

2 Vertex Coloring is NP-Complete

Problem

Can the vertices of G be colored with at most k colors so that adjacent vertices have different colors?

Hard Core

$\boxed{3\text{-CNF-SAT} \leq_p 3\text{-COLOR}}$

Then 3-COLOR is the special case $k = 3$ of VERTEX-COLOR.

Actual Proof

In NP. Certificate = one color per vertex. Check every edge; its endpoints must have different colors.

Build. From a 3-CNF formula:

- Create special vertices T, F, B and connect them as a triangle. They must get three different colors.
- For each variable x_i , create x_i and $\neg x_i$; connect them to each other and to B . Thus they receive opposite T/F colors.
- For each clause $(\ell_1 \vee \ell_2 \vee \ell_3)$, attach the standard clause gadget to its three literal vertices. The gadget is 3-colorable exactly when at least one literal has color T .

$\Rightarrow$. If the formula is satisfiable, color each literal vertex from its truth value. Every clause has a true literal, so every clause gadget can be 3-colored. Hence the whole graph is 3-colorable.

$\Leftarrow$. If the graph has a 3-coloring, each pair $x_i, \neg x_i$ gets opposite T/F colors, giving a truth assignment. Every clause gadget forces at least one of its literals to be T , so every clause is satisfied.

Time. Each variable and clause adds only a constant-size gadget, so the graph size is polynomial.

Conclusion. 3-COLOR is NP-complete. Since it is a special case of VERTEX-COLOR,

VERTEX-COLOR is NP-complete.

What to Draw if Asked

Draw the T, F, B triangle, the variable pair $x_i, \neg x_i$, and your course's standard 3-SAT clause gadget.

3 Clique Partition (Graph Partition) is NP-Complete

Problem

Can $V(G)$ be partitioned into at most k cliques?

Reduction

$\text{VERTEX-COLOR} \leq_p \text{CLIQUE-PARTITION}$

Given (G, k) , output $(\overline{G}, k)$.

Actual Proof

In NP. Certificate = a partition into at most k groups. Check that every pair inside each group is adjacent.

Build. Construct the complement graph $\overline{G}$.

$\Rightarrow$. If G has a proper k -coloring, each color class is an independent set in G . Therefore each color class becomes a clique in $\overline{G}$. So $\overline{G}$ can be partitioned into at most k cliques.

$\Leftarrow$. If $\overline{G}$ is partitioned into at most k cliques, each part is an independent set in G . Give one color to each part. No adjacent vertices of G receive the same color, so G is k -colorable.

Time. Constructing $\overline{G}$ takes $O(|V|^2)$.

Conclusion.

CLIQUE-PARTITION is NP-complete.

4 Edge Coloring is NP-Complete

Problem

Can the edges of G be colored with at most k colors so that edges sharing a vertex have different colors?

Known NP-Complete Source

Holyer's theorem: deciding whether a *cubic* graph is 3-edge-colorable is NP-complete.

Actual Proof

In NP. Certificate = one color per edge. For each vertex, check that all incident edges have different colors.

Build. Take an instance G of cubic 3-edge-coloring and output the general edge-coloring instance $(G, 3)$.

$\Rightarrow$. If the cubic graph G is 3-edge-colorable, then $(G, 3)$ is clearly a YES-instance of EDGE-COLOR.

$\Leftarrow$. If $(G, 3)$ is a YES-instance of EDGE-COLOR, then the same coloring is a valid 3-edge-coloring of the original cubic graph.

Time. The reduction only copies G and writes $k = 3$, so it is polynomial.

Conclusion. By Holyer's NP-completeness result and membership in NP,

EDGE-COLOR is NP-complete.

5 Graph Isomorphism: In NP, Not Known NP-Complete

Problem

Are G_1 and G_2 the same graph up to a renaming of vertices?

Proof that Graph Isomorphism is in NP

Certificate. A bijection $f : V(G_1) \rightarrow V(G_2)$.

Verify. Check that f is one-to-one and onto. Then for every pair u, v , check

$$(u, v) \in E(G_1) \iff (f(u), f(v)) \in E(G_2).$$

There are at most $O(|V|^2)$ pairs, so verification is polynomial.

Conclusion.

GRAPH-ISOMORPHISM $\in$ NP.

Important

Ordinary GRAPH-ISOMORPHISM is **not known to be NP-complete**. Do not write an NP-completeness proof for it unless the question is actually about SUBGRAPH-ISOMORPHISM.

6 Subgraph Isomorphism is NP-Complete

Problem

Is G_1 isomorphic to some subgraph of G_2 ?

Reduction

$$\text{CLIQUE} \leq_p \text{SUBGRAPH-ISOMORPHISM}$$

Given (G, k) , set $G_1 = K_k$ and $G_2 = G$.

Actual Proof

In NP. Certificate = an injective mapping from vertices of G_1 to vertices of G_2 . Check that every edge of G_1 maps to an edge of G_2 .

Build. From (G, k) , construct the complete graph K_k and output (K_k, G) .

$\Rightarrow$. If G has a clique of size k , the subgraph induced by those k vertices contains a copy of K_k . Hence K_k is isomorphic to a subgraph of G .

$\Leftarrow$. If K_k is isomorphic to a subgraph of G , the corresponding k vertices of G are pairwise adjacent. Therefore they form a clique of size k .

Time. Building K_k takes $O(k^2)$, so the reduction is polynomial.

Conclusion.

SUBGRAPH-ISOMORPHISM is NP-complete.